Year 10 Mathematics

Graphs 2

8.1 Straight-Line Conversion Graphs

Straight-Line Conversion Graphs

STARTER · 5 MIN

Date: _____ **Title:** Straight-Line Conversion Graphs

LEARNING OBJECTIVES

- Read values from a conversion graph
- Draw a conversion graph from a table of values
- Use conversion graphs to solve problems

STARTER *Answer in your book. No calculators.*

1. Work out 40 × 1.2
2. Work out 90 ÷ 1.2
3. 5 miles = 8 km. How many km is 10 miles?
4. Work out 1.8 × 50 + 32

Starter — Answers

STARTER

Tick or correct in green pen

1. $40 \times 1.2 = \mathbf{48}$
2. $90 \div 1.2 = \mathbf{75}$
3. $10 \text{ miles} = 2 \times 8 = \mathbf{16 \text{ km}}$
4. $90 + 32 = \mathbf{122}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can read values from a conversion graph	☐
SILVER	I can draw a conversion graph from a table of values	☐
GOLD	I can use conversion graphs to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Conversion Graphs

INTRODUCTION · 3 MIN

A conversion graph changes one unit into another.

Reading across

Go up from the x-axis to the line, then across.

£40 → €48

Reading back

Go across from the y-axis to the line, then down.

€90 → £75

Drawing

Make a table, plot points, join with a ruler.

0 miles = 0 km

Large values

Read a smaller value and multiply.

£300 = 3 × £100

Examples 1 and 2 — Reading a Conversion Graph

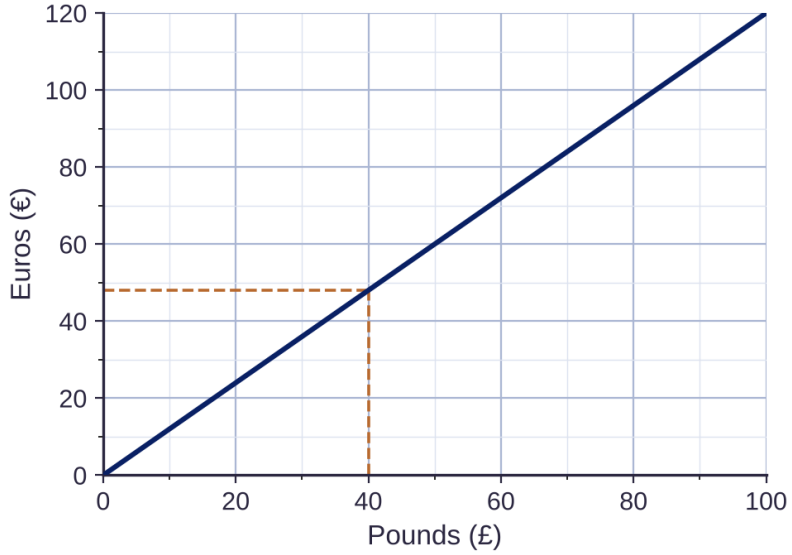
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

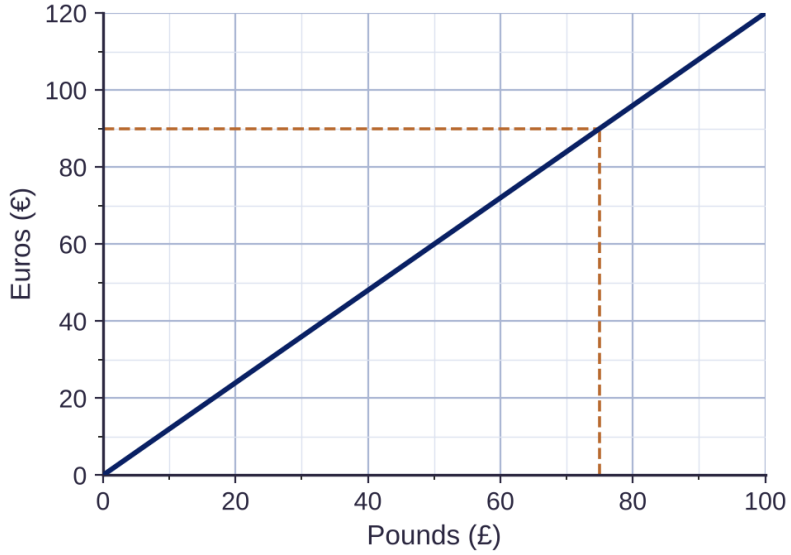
Example 1

Use the graph to change £40 into euros.



Example 2

Use the graph to change €90 into pounds.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Go up from £40 to the line

Go across to the euros axis

£40 = €48

Tip: use a ruler and draw dashed lines.

Example 2

Go across from €90 to the line

Go down to the pounds axis

€90 = £75

Tip: start on the axis of the unit you have.

Task 1

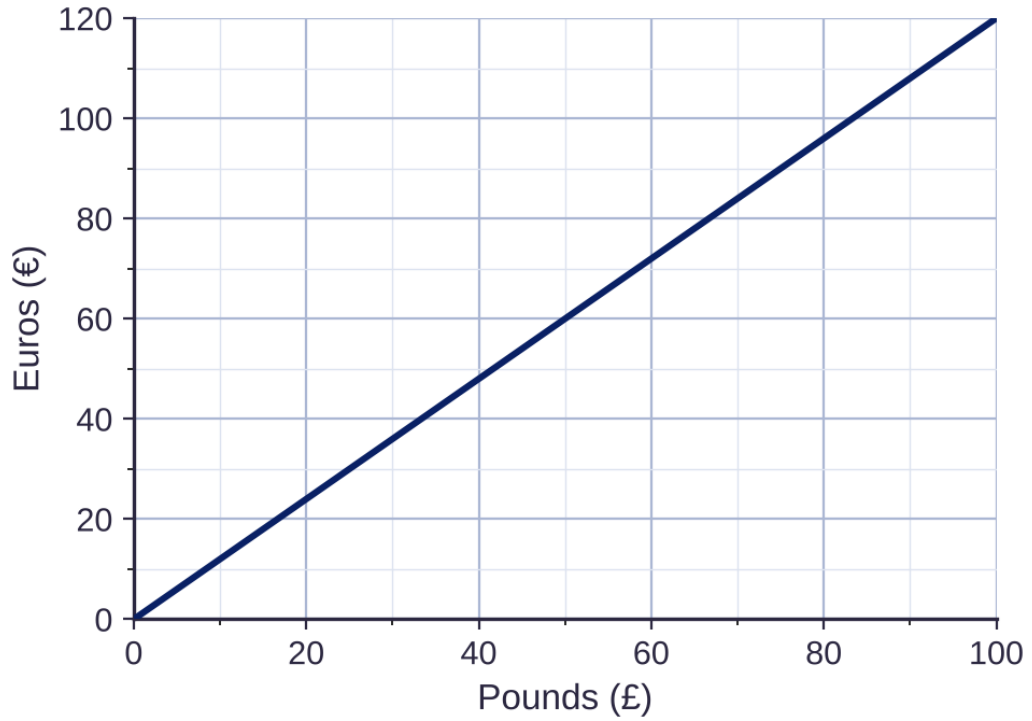
PRACTICE · 9 MIN

BRONZE

Use the graph to answer each question

1. Change £20 into euros
2. Change £70 into euros
3. Change €60 into pounds
4. Change €108 into pounds
5. Change £300 into euros

★ How many euros do you get for £1?



Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. £20 = **€24**
2. £70 = **€84**
3. €60 = **£50**
4. €108 = **£90**
5. £100 = €120, £300 = 3 × 120 = **€360**

★ €120 ÷ 100 = **€1.20**

Examples 3 and 4 — Drawing a Conversion Graph

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

5 miles = 8 km.

Draw a graph to convert between

miles and kilometres for 0 to 50 miles.

Example 4

Use the miles–kilometres graph

to change 64 km into miles.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

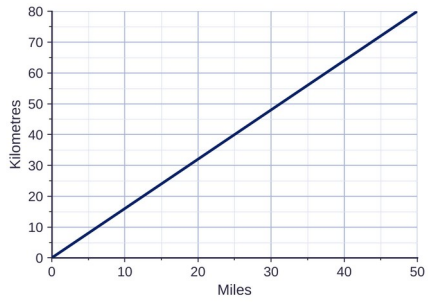
Example 3

0 miles = 0 km

25 miles = 40 km

50 miles = 80 km

Plot and join with a ruler



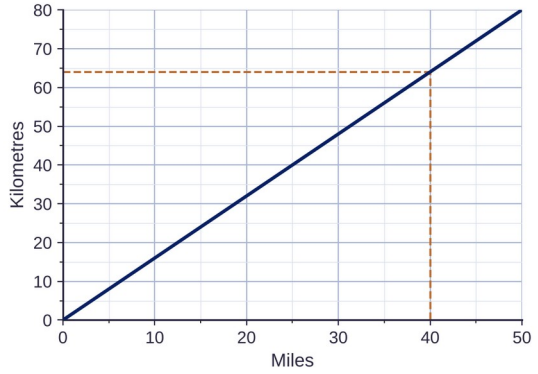
Tip: a straight line through (0, 0).

Example 4

Go across from 64 km to the line

Go down to the miles axis

64 km **= 40 miles**



Tip: check: 8 × 8 = 64 and 5 × 8 = 40.

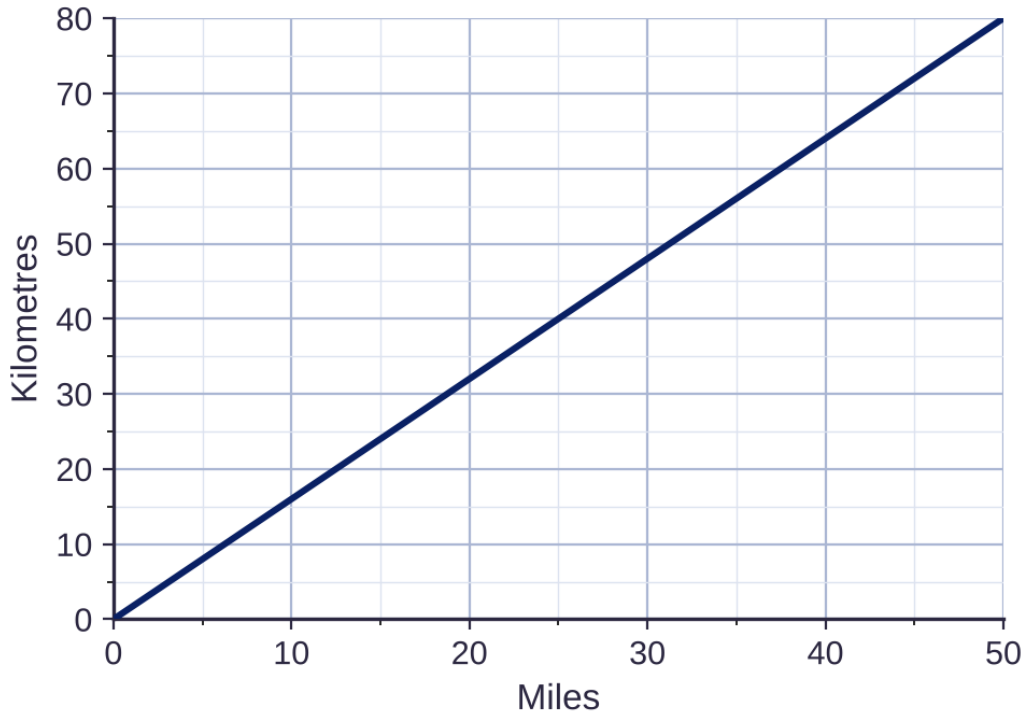
Task 2

PRACTICE · 9 MIN

SILVER

Use the graph to answer each question

1. Change 30 miles into km
2. Change 20 km into miles
3. Change 45 miles into km
4. Change 72 km into miles
5. A road is 150 miles long. How many km is this?



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 30 miles = **48 km**

2. 20 km = **12.5 miles**

3. 45 miles = **72 km**

4. 72 km = **45 miles**

5. 50 miles = 80 km, 150 miles = 3 × 80 = **240 km**

Examples 5 and 6 — Graphs Not Through the Origin

EXAMPLES · 5 MIN

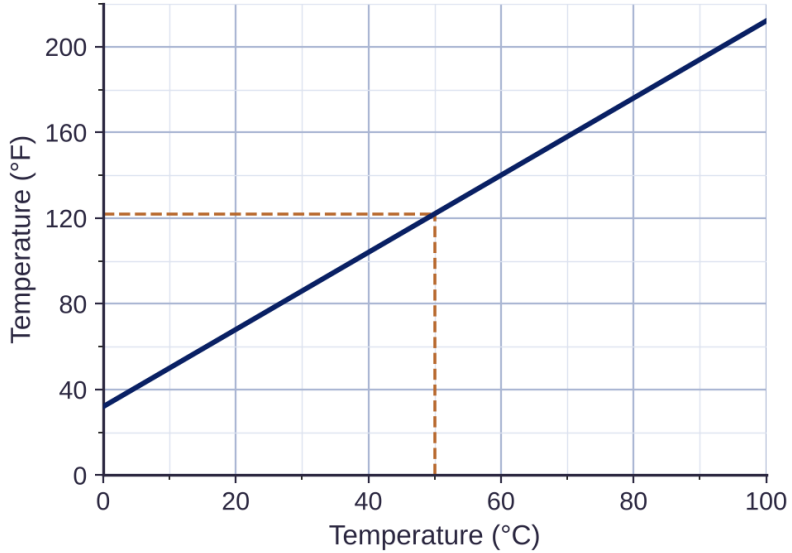
GOLD

Exam-style: show all your working

Example 5

(1 mark)

Use the graph to change 50 °C into °F.



Example 6

(2 marks)

Water boils at 212 °F.

Use the graph to find this in °C.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Go up from 50 °C to the line

Go across to the °F axis

50 °C **= 122 °F**

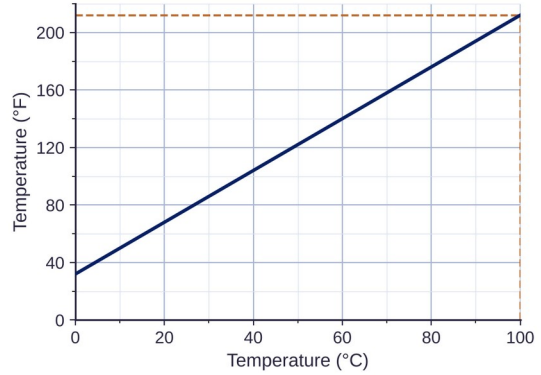
Tip: this line does not start at (0, 0).

Example 6

Go across from 212 °F to the line

Go down to the °C axis

212 °F **= 100 °C**



Tip: read the scale carefully: each small square is 20 °F.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

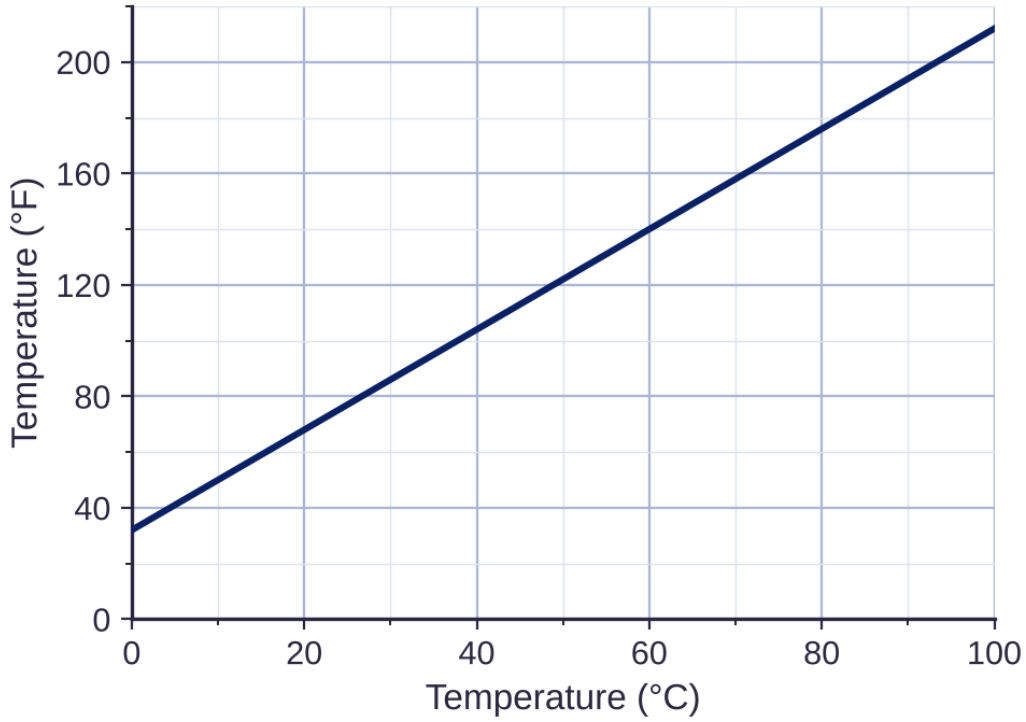
1. Change 20 °C into °F (1 mark)

2. Change 30 °C into °F (1 mark)

3. Change 140 °F into °C (1 mark)

4. What temperature in °F is 0 °C? (1 mark)

5. Which is hotter, 40 °C or 100 °F? Explain. (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. 20 °C = **68 °F**
2. 30 °C = **86 °F**
3. 140 °F = **60 °C**
4. 0 °C = **32 °F**
5. 40 °C = 104 °F, **so 40 °C is hotter**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Use £1 = €1.20. Change £50 into euros.

SILVER Use 5 miles = 8 km. Change 40 km into miles.

GOLD Use the temperature graph. Change 10 °C into °F.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $50 \times 1.2 = \mathbf{€60}$

SILVER $40 \div 8 \times 5 = \mathbf{25\ miles}$

GOLD $10\ °C = \mathbf{50\ °F}$

Exam Practice — Reading Conversion Graphs

EXAM PRACTICE · 10 MIN

BRONZE

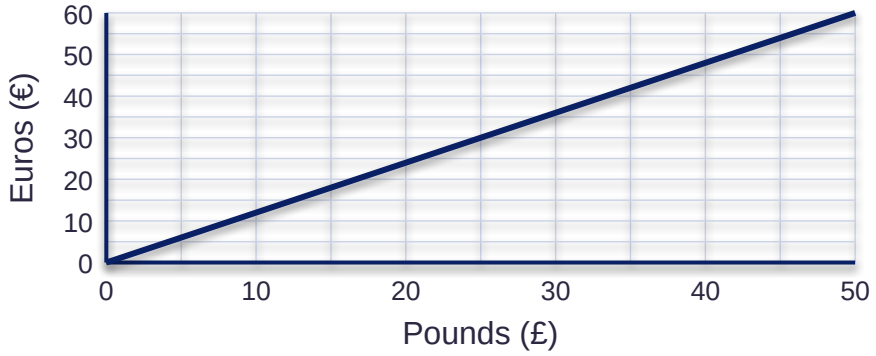
IGCSE exam-style: answer on the line, show all your working

Question 1

(1 mark)

The graph converts between pounds (£) and euros (€).

Use the graph to change £30 into euros.



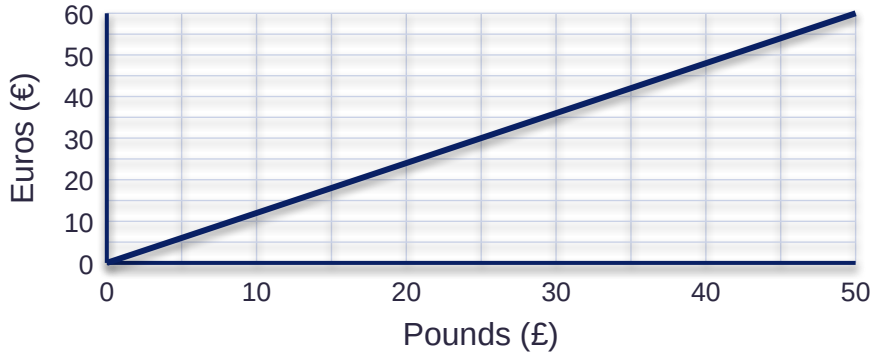
..... €

Question 2

(1 mark)

The graph converts between pounds (£) and euros (€).

Use the graph to change €54 into pounds.



..... £

Exam Practice — Drawing Conversion Graphs

EXAM PRACTICE · 10 MIN

SILVER

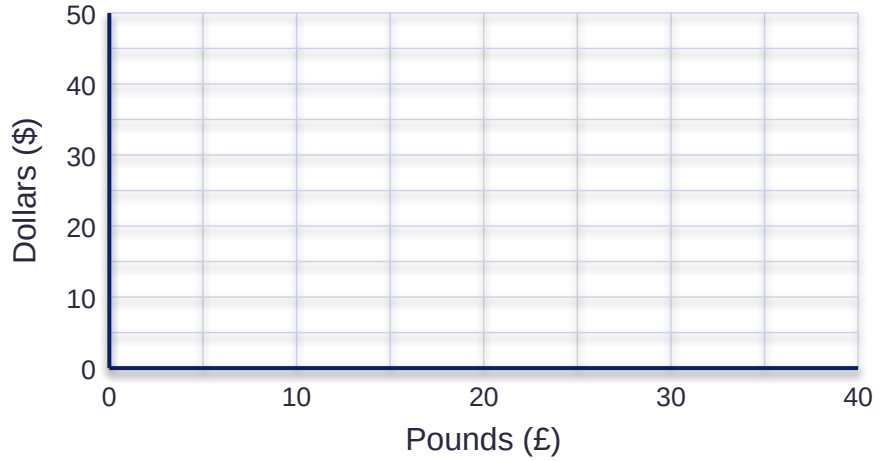
IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

£1 = $1.25

On the grid, draw a conversion graph for £0 to £40.



Question 4

(2 marks)

Kai has £38. A jacket costs $45.

Use your graph from Question 3 to decide

if Kai has enough money.

Show how you decide.

.....

Exam Practice — Graphs Not Through the Origin

EXAM PRACTICE · 10 MIN

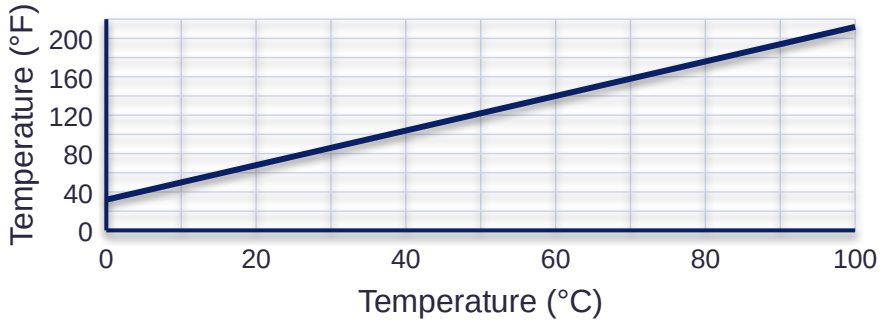
GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(2 marks)

The graph converts between °C and °F.



(a) Change 40 °C into °F. (1)

..... °F

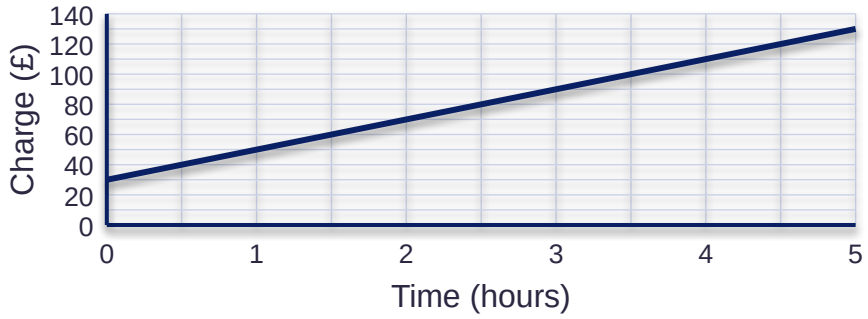
(b) Change 140 °F into °C. (1)

..... °C

Question 6

(2 marks)

The graph shows a plumber's charges.



(a) Write down the call-out fee. (1)

..... £

(b) Work out the hourly rate. (1)

..... £

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

Up from £30, across to €-axis

€36 B1

Tip: draw your lines on the graph.

Question 2

Across from €54, down to £-axis

£45 B1

Tip: accept answers within half a small square.

Exam Practice — Mark Scheme

CORRECTIONS

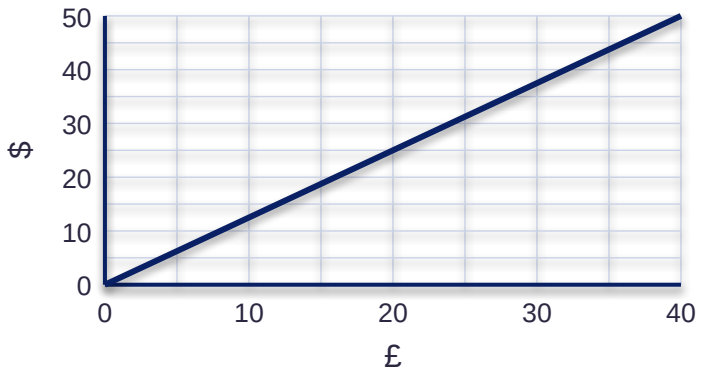
SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

e.g. £40 = $50, £0 = $0 — **M1**

Ruled line (0, 0) to (40, 50) — **A1**



Tip: find one point, e.g. £40 = $50.

Question 4

£38 = $47.50 (from graph) — **M1**

Yes: $47.50 > $45 — **A1**

Tip: or convert $45 to £36, which is less than £38.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 104 °F B1

(b) 60 °C B1

Tip: the line starts at 32 °F, not at 0.

Question 6

(a) £30 B1

(b) (130 − 30) ÷ 5

= £20 per hour B1

Tip: hourly rate = gradient of the line.